

Connect your mobile to the same network of the server over Wi-Fi or by using USB tethering, and test the resources by entering the server IP in these apps.

Bridging with HTTP

Since CoAP supports a subset of the REST model similar to HTTP, it is easy to integrate both protocols using proxy endpoints. An HTTP-CoAP proxy is used to request resources on a CoAP server from the HTTP client and reverse proxy is used to request resources on the HTTP server from CoAP clients. Eclipse Californium comes with a proxy library and an example *cf-proxy* for this.

By running *cf-proxy*, you can talk to the CoAP server from the HTTP client as follows: `http://hostname:8080/proxy/coap://hostname:5683/resource`, where *hostname* represents the proxy and *hostname* represents the origin server.

Another implementation is the npm package *ponte*, a current Eclipse IoT project that bridges MQTT, CoAP and HTTP protocols. To try this, install the *ponte*, *bunyan* packages using npm and run the following command:

```
ponte -v | bunyan
```

Using this bridge, identical URIs for the resource named *abcd* are `coap://hostname:5683/r/abcd` and `http://`

`hostname:3000/resources/abcd`

Public servers and cloud platforms

Compared to other protocols, currently, very few cloud platforms support CoAP like *thethings.io*, *ThingMQ*, etc. A few public servers with test resources to get started with clients are listed on `coap://californium.eclipse.org/` or `coap://vs0.inf.ethz.ch/`, `coap://coap.me/`.

I hope this article helps you to bootstrap the CoAP protocol, and its tools and libraries. Please follow the given pointers for detailed coverage of the protocol and more hands-on information. 

References

- [1] <https://tools.ietf.org/rfc/rfc7252> [RFC7252]
- [2] 'CoAP: The Web of Things Protocol', Zach Shelby, ARM IoT Tutorial.
- [3] 'Hands-on with CoAP', Matthias Kovatsch and Julien Vermillard, EclipseCon France 2014
- [4] github.com/eclipse/californium.git

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